

Where the marks are

Ten systems carry the DOH IM specialist paper. Study them in this order, not in textbook order — the top four are 52% of your score, and all of Dermatology is worth a third of Cardiology.

60% Pass mark you are aiming past	52% Sits in the top four systems	90 Q Roughly what 60% of 150 looks like
Cardiology ACS pathway & reperfusion windows · AF rate vs rhythm + CHA₂DS₂-VASc · HFrEF four pillars · murmurs · ECG blocks and axis		18% ≈27 Q
Endocrinology T2DM agent choice by comorbidity · DKA vs HHS · thyroid function patterns · adrenal insufficiency · osteoporosis · Cushing / phaeo screening		12% ≈18 Q
Respiratory Medicine Asthma vs COPD escalation ladders · CURB-65 and CAP antibiotics · PE probability scoring · pleural effusion (Light's) · O₂ targets		12% ≈18 Q
Nephrology AKI staging and pre-renal vs ATN · CKD staging and referral · hyperkalaemia sequence · acid–base with anion gap · nephrotic vs nephritic		10% ≈15 Q
Gastroenterology Upper GI bleed and variceal care · decompensated cirrhosis (SBP, HE, ascites) · IBD flares · coeliac serology · LFT patterns · pancreatitis		10% ≈15 Q
Rheumatology Acute gout vs septic arthritis · SLE and antibody patterns · giant cell arteritis · vasculitis (ANCA) · seronegative spondyloarthritis		8% ≈12 Q
Neurology Stroke thrombolysis window vs thrombectomy · SAH work-up · seizure and status ladder · MS and GBS · headache red flags · neuropathy		8% ≈12 Q
Haematology Anaemia by MCV and iron studies · transfusion thresholds · VTE and anticoagulation choice · myeloma triad · leukaemia and lymphoma basics		8% ≈12 Q
Infectious Disease Sepsis bundle timing · TB diagnosis and regimen · HIV screening and prophylaxis · endocarditis (Duke) · meningitis · travel fever		8% ≈12 Q
Dermatology Drug eruptions and SJS/TEN · cellulitis vs necrotising fasciitis · psoriasis · skin signs of internal disease · pigmented lesion red flags		6% ≈9 Q

Six weeks out

Heaviest systems first, while your attention is cheapest. The last week buys nothing new — it only converts what you already half-know into marks.

THE DAILY RHYTHM — SAME EVERY DAY, ALL SIX WEEKS

40 questions

Timed block in the week's system. Timed from day one.

Full review

Every item — including the ones you guessed right.

Leak list

One line per miss: stem clue, your pick, the actual rule.

15 min recall

Flashcards on last week's system, not this week's.

W-6

18% OF PAPER

Baseline & Cardiology

Sit a full diagnostic on day one — before you revise anything. You need a real weakness map, not a felt one.

ACS pathway and reperfusion windows · AF rate vs rhythm, CHA₂DS₂-VASc, DOAC vs warfarin · HFrEF four pillars · murmur-to-lesion mapping · ECG blocks.

W-5

24% OF PAPER

Endocrine & Respiratory

The two biggest blocks after Cardiology. Both are algorithm-heavy, which means they are learnable fast.

T2DM agent choice by comorbidity · DKA vs HHS criteria and fluid order · thyroid patterns · adrenal insufficiency · asthma and COPD ladders · CURB-65 · PE probability.

W-4

20% OF PAPER

Nephrology & Gastroenterology

Where most candidates leak marks on numbers rather than concepts. Learn the cutoffs cold.

AKI staging, pre-renal vs ATN urinary indices · hyperkalaemia sequence · anion-gap acid-base · decompensated cirrhosis (SBP, HE, ascites) · upper GI bleed · IBD flare.

W-3

16% OF PAPER

Rheumatology & Neurology

Pattern-recognition systems. Drill stems, not textbooks — the marks are in the clue-to-diagnosis jump.

Gout vs septic arthritis · SLE and antibody patterns · giant cell arteritis (treat before biopsy) · ANCA vasculitis · stroke windows · status epilepticus ladder · headache red flags.

W-2

22% OF PAPER

Haematology, ID & Dermatology

Three lighter systems together. Derm is only 6% — cap it at two days and move on.

Anaemia by MCV and iron studies · transfusion thresholds · VTE choice and duration · sepsis bundle timing · TB and HIV basics · endocarditis · drug eruptions and SJS/TEN.

W-1

NO NEW TOPICS

Mocks and leaks only

Two full timed mocks, spaced. Everything between them is rebuilt from your own wrong answers.

Days 7 and 4: full mock, full review. Days 6, 5, 3: leak list plus the cram sheets in this pack. Day 2: algorithm sheets out loud. Day before: one light 20-Q block, then stop.

RIGHT ANSWER, WRONG REASON

A guess you got right is a miss that hasn't happened yet. Mark it in the leak list the same as a wrong answer.

NEVER FINISH ON A MISS

Got one wrong at the end of a block? Do five more of that subtopic before you close the laptop. Fewer weeks means shorter blocks, same order.

Cutoffs and thresholds

The values a stem hangs its answer on. If you can't produce the number, you can't rule out the distractor built around it.

CARDIOLOGY

Hypertension, clinic / ABPM daytime	≥140/90 · ≥135/85
LDL target, established ASCVD	<1.4 mmol/L + ≥50% drop
LDL target, high risk	<1.8 mmol/L
STEMI — primary PCI from first contact	≤90 min
Lyse instead if PCI unavailable within	120 min · door-to-needle ≤30
Reperfusion window from symptom onset	≤12 h
HFrEF / HFmrEF / HFpEF by LVEF	≤40 · 41–49 · ≥50%
Anticoagulate AF — CHA ₂ DS ₂ -VASc	≥2 men · ≥3 women

ENDOCRINE

Diabetes — HbA1c / FPG / 2-h OGTT	≥6.5% · ≥7.0 · ≥11.1
Prediabetes — HbA1c / fasting	5.7–6.4% · 5.6–6.9
DKA triad	Glucose >13.9 · pH <7.3 · HCO ₃ <18 · ketones ≥3.0
HHS	Glucose >33.3 · osm >320 · pH >7.3 · ketones low
Hypoglycaemia — alert / significant	<3.9 · <3.0 mmol/L
Treat subclinical hypothyroid when TSH	>10 mIU/L
Adrenal insufficiency — 9 am cortisol	<100 likely · >400 nmol/L excludes

RESPIRATORY

SpO ₂ target — usual / CO ₂ retention risk	94–98% · 88–92%
CURB-65 — home / admit / ICU review	0–1 · 2 · ≥3
Acute severe vs life-threatening asthma	PEF 33–50% · <33% or SpO ₂ <92%
Type 2 respiratory failure — PaCO ₂	>6.0 kPa (45 mmHg)
NIV in COPD when pH	<7.35 with PaCO ₂ >6.0
Light's — exudate if any one	Pr ratio >0.5 · LDH ratio >0.6 · LDH >⅔ ULN
Wells — PE likely	>4 points

NEUROLOGY

IV thrombolysis window	≤4.5 h from onset
Thrombectomy, anterior LVO	≤6 h · to 24 h if imaging-selected
BP ceiling before thrombolysis	<185/110 mmHg
Status epilepticus starts at	5 min of seizure
LP for xanthochromia — earliest	12 h after headache onset

RENAL & ELECTROLYTES

AKI stage 1 — creatinine	×1.5–1.9 in 7 d or ↑≥26.5 μmol/L in 48 h
AKI stage 1 — urine output	<0.5 mL/kg/h for 6–12 h
AKI stage 2 / stage 3	×2.0–2.9 · ×3.0 or ≥353.6 or RRT
Pre-renal vs ATN — FeNa	<1% · >2%
CKD G3a / G3b / G4 / G5 — eGFR	45–59 · 30–44 · 15–29 · <15
Hyperkalaemia — emergency at	≥6.5 or any ECG change
Max Na correction in 24 h	≤8–10 mmol/L
Corrected Ca (mmol/L)	measured + 0.02 × (40 – albumin)
Normal anion gap	8–12 mmol/L

HAEMATOLOGY & VTE

Transfuse — stable / ACS	Hb <70 · <80 g/L
Platelets — prophylactic / procedure	<10 · <50 ×10 ⁹ /L
MCV — micro / normo / macro	<80 · 80–100 · >100 fL
Iron deficiency — ferritin	<30 μg/L (<100 if inflamed)
Anticoagulation — provoked VTE	3 months
INR target — usual / mechanical mitral	2.5 · 3.0
Myeloma CRAB	Ca >2.75 · Cr >177 · Hb <100 · lytic bone

SEPSIS & INFECTION

qSOFA ≥2 of	RR ≥22 · altered mentation · SBP ≤100
Antibiotics from recognition	within 1 h
Fluid if hypotensive or lactate ≥4	30 mL/kg crystalloid
Septic shock	Pressors for MAP ≥65 + lactate >2
Neutropenia / severe — ANC	<1.0 · <0.5 ×10 ⁹ /L
Neutropenic fever	≥38.3 °C once or ≥38.0 for 1 h
SBP — ascitic neutrophils	≥250 /mm ³

LIVER & GI

Glasgow-Blatchford — consider discharge	0–1
Endoscopy in upper GI bleed	≤24 h · ≤12 h if variceal/unstable
SAAG — portal hypertension	≥11 g/L
AST:ALT suggesting alcohol	>2:1
Severe alcoholic hepatitis — Maddrey	≥32

Drug of choice, and what changes it

Most management stems are decided by the third column — the comorbidity, allergy or contraindication that moves the answer off the default.

CARDIOVASCULAR

SITUATION	FIRST LINE	WHAT MOVES THE ANSWER
Hypertension, under 55, not Black African/Caribbean	ACE inhibitor or ARB	Age ≥55 or Black African/Caribbean → CCB first
Hypertension with diabetes or proteinuria	ACE inhibitor or ARB, any age	Pregnancy → stop; labetalol, nifedipine or methyldopa
Heart failure, reduced EF	ARNI (or ACEi) + beta-blocker + MRA + SGLT2i	Loop diuretic treats congestion, not mortality
Acute coronary syndrome	Aspirin 300 mg + P2Y12 + anticoagulant + high-intensity statin	Nitrate and morphine are symptomatic only
AF — stroke prevention	DOAC	Mechanical valve or moderate–severe mitral stenosis → warfarin

ENDOCRINE

SITUATION	FIRST LINE	WHAT MOVES THE ANSWER
Type 2 diabetes, no complications	Metformin + lifestyle	Stop below eGFR 30; hold in acute illness or contrast
Type 2 diabetes with ASCVD, HF or CKD	SGLT2 inhibitor, regardless of HbA1c	Euglycaemic DKA — glucose can be near normal
Type 2 diabetes with obesity	GLP-1 receptor agonist	Avoid with medullary thyroid cancer or MEN2
Diabetic ketoacidosis	IV fluids first, then fixed-rate insulin 0.1 u/kg/h	K ⁺ below 3.5 → replace before starting insulin
Thyroid storm	Propylthiouracil or carbimazole + beta-blocker + steroid	Iodine only ≥1 h after the thionamide
Adrenal crisis	IV hydrocortisone 100 mg + fluids	Treat first — never wait for the synacthen result

RESPIRATORY & INFECTION

SITUATION	FIRST LINE	WHAT MOVES THE ANSWER
Asthma maintenance	ICS-formoterol as MART	SABA alone is never the correct maintenance answer
COPD maintenance	LABA + LAMA	Add ICS if eosinophils high or exacerbations recur
Community-acquired pneumonia	Amoxicillin if low severity; co-amoxiclav + macrolide if moderate–severe	Penicillin allergy or atypical picture → macrolide
Clostridioides difficile	Oral vancomycin or fidaxomicin	Stop the culprit antibiotic; loperamide is wrong
Tuberculosis, new case	RIPE 2 months, then rifampicin + isoniazid 4 months	Ethambutol → baseline visual acuity; isoniazid → pyridoxine

NEUROLOGY & RHEUMATOLOGY

SITUATION	FIRST LINE	WHAT MOVES THE ANSWER
Status epilepticus	IV lorazepam, repeat once	Then levetiracetam, valproate or phenytoin; then anaesthesia
Acute gout	NSAID, colchicine or steroid	Continue allopurinol if already on it; never start it mid-flare
Giant cell arteritis	High-dose prednisolone immediately	Visual loss → IV methylprednisolone; biopsy after starting
SLE	Hydroxychloroquine for everyone	Organ involvement → steroid plus immunosuppressant

EMERGENCIES & ANTIDOTES

SITUATION	FIRST LINE	WHAT MOVES THE ANSWER
Anaphylaxis	IM adrenaline 0.5 mg of 1:1000, anterolateral thigh	Repeat at 5 min; steroid and antihistamine are adjuncts
Hyperkalaemia with ECG changes	IV calcium gluconate	Calcium protects the myocardium — it does not lower K ⁺
Paracetamol overdose	N-acetylcysteine	Staggered or unknown timing → treat without the nomogram
Warfarin with major bleeding	IV vitamin K + prothrombin complex concentrate	FFP only when PCC is unavailable
Beta-blocker or calcium-channel blocker overdose	Glucagon, or high-dose insulin euglycaemic therapy	Add IV calcium for the calcium-channel blocker

The clue, the call, the next move

Written examiners plant one phrase that settles the diagnosis. Learn the phrase and the immediate next step — the option list is usually four plausible investigations and one right order.

CARDIOVASCULAR & RESPIRATORY

PHRASE IN THE STEM	THINK	NEXT MOVE
Tearing chest pain radiating to the back, blood pressure different in each arm	Aortic dissection	CT aortogram. Control heart rate with IV labetalol before any vasodilator.
Pleuritic pain and sudden breathlessness after immobility, surgery or a long flight	Pulmonary embolism	Wells score, then CTPA. Anticoagulate while waiting if probability is high.
Raised JVP, muffled heart sounds, hypotension, pulsus paradoxus	Cardiac tamponade	Urgent bedside echo, then pericardiocentesis.
Syncope on exertion with an ejection systolic murmur radiating to the carotids	Severe aortic stenosis	Echo. Avoid nitrates and other vasodilators.
COPD patient becomes drowsy after starting high-flow oxygen	Hypercapnic respiratory failure	ABG. Titrate O ₂ to 88–92%; NIV if pH is below 7.35.

NEUROLOGY

PHRASE IN THE STEM	THINK	NEXT MOVE
Thunderclap headache, worst of their life, CT head reported normal	Subarachnoid haemorrhage	Lumbar puncture at 12 h or later for xanthochromia.
Sudden loss of vision in one eye, jaw claudication, scalp tenderness, high ESR	Giant cell arteritis	High-dose steroid immediately — biopsy afterwards, never before.
Back pain with leg weakness or bladder change in a patient with known cancer	Metastatic cord compression	Whole-spine MRI urgently plus dexamethasone. Hours matter.
Ascending weakness with absent reflexes days after a diarrhoeal illness	Guillain-Barré syndrome	Serial FVC — not oxygen saturation. IVIG or plasma exchange.
Headache worse lying flat, morning vomiting, papilloedema	Raised intracranial pressure	Imaging before lumbar puncture, always.

ENDOCRINE & METABOLIC

PHRASE IN THE STEM	THINK	NEXT MOVE
Euvolaemic hyponatraemia, urine osmolality above 100, urine sodium above 30	SIADH	Fluid restriction, then hunt the cause: drug, lung, CNS or malignancy.
Fatigue, skin pigmentation, low sodium, high potassium, postural drop	Primary adrenal insufficiency	9 am cortisol and short synacthen — but give hydrocortisone first if unwell.
Episodic headache, sweating and palpitations with labile blood pressure	Phaeochromocytoma	Plasma or 24 h urinary metanephrines. Alpha-blockade before beta-blockade.
Acidotic and unwell on an SGLT2 inhibitor with near-normal glucose	Euglycaemic DKA	Check ketones and ABG. Treat as DKA — the glucose reassures nobody.
Fever, agitation and fast AF in a patient with Graves after an infection	Thyroid storm	Thionamide, then beta-blocker and steroid. Iodine only 1 h after the thionamide.

GASTRO, INFECTION, RHEUMATOLOGY, SKIN

PHRASE IN THE STEM	THINK	NEXT MOVE
Cirrhosis with ascites who becomes febrile, confused or tender	Spontaneous bacterial peritonitis	Diagnostic tap. Neutrophils $\geq 250/\text{mm}^3$ → antibiotics plus albumin.
Profuse watery diarrhoea days into a course of antibiotics	Clostridioides difficile	Stool toxin, oral vancomycin, stop the culprit. Never an antimotility drug.
One hot, swollen, exquisitely painful joint with fever	Septic arthritis until proven otherwise	Aspirate the joint before the first dose of antibiotic.
Target lesions with mouth and eye ulceration after starting a new drug	Stevens-Johnson / TEN	Stop the drug immediately; burns-unit level supportive care.
Haemoptysis with haematuria and a rising creatinine	Pulmonary-renal syndrome	ANCA and anti-GBM, urgent renal referral. Do not wait for the biopsy.

Chest pain, fast heart, breathless

Management-order questions live here. The mark is for the step that comes *next* — not the step that is most thorough.

Acute coronary syndrome

Ischaemic pain with diaphoresis, nausea or radiation. The ECG and the troponin timing decide everything else.

- 1 ECG within 10 minutes of arrival**
Repeat if pain continues and the first is non-diagnostic. ST depression in V1–V3 → posterior leads before calling it NSTEMI.
- 2 Aspirin 300 mg, a P2Y12 inhibitor and an anticoagulant**
Nitrate and morphine relieve symptoms only. Oxygen just for hypoxia.
- 3 STEMI → primary PCI within 90 minutes of first medical contact**
Not achievable inside 120 minutes → fibrinolyse with a door-to-needle time of 30 minutes or less, then transfer.
- 4 NSTEMI or unstable angina → risk-stratify, then time the angiogram**
High GRACE risk or ongoing ischaemia → angiography within 24 hours. Everyone leaves on a high-intensity statin, DAPT, ACE inhibitor, beta-blocker and an echo.

TRAP A single normal troponin never excludes ACS. When every option is a sensible test, the answer is usually the repeat troponin at the right interval.

Atrial fibrillation with a fast ventricular rate

Irregularly irregular pulse above 110. The first question is haemodynamic, never pharmacological.

- 1 Unstable? Synchronised DC cardioversion now**
Shock, syncope, myocardial ischaemia or acute heart failure — do not rate-control your way through it.
- 2 Stable → rate control with a beta-blocker or diltiazem**
Digoxin suits the sedentary patient or heart failure; amiodarone if rhythm control is intended.
- 3 Time the onset before considering cardioversion**
Clearly under 48 hours → cardioversion is on the table. 48 hours or unknown → three weeks of anticoagulation, or a TOE first.
- 4 Score the stroke risk separately, then find the driver**
CHA₂DS₂-VASc ≥2 men or ≥3 women → DOAC; mechanical valve or moderate–severe mitral stenosis → warfarin. Look for sepsis, thyrotoxicosis, PE or alcohol.

TRAP Never pair verapamil or diltiazem with a beta-blocker, and avoid both in decompensated heart failure with reduced ejection fraction.

Suspected pulmonary embolism

Pleuritic pain, breathlessness, tachycardia — usually with immobility, surgery, malignancy or pregnancy in the stem.

- 1 Score the probability first**
Wells: PE unlikely → D-dimer. PE likely → straight to CTPA, no D-dimer.
- 2 Unstable → bedside echo for right ventricular strain**
Peri-arrest or persistent hypotension with a convincing story → thrombolyse without waiting for the CTPA.
- 3 Anticoagulate while you wait when probability is high**
Unless active bleeding or a clear contraindication makes the risk unacceptable.
- 4 Confirmed → DOAC, then set the duration before discharge**
Severe renal impairment or antiphospholipid syndrome → LMWH into warfarin. Provoked 3 months; unprovoked or active cancer at least 3, extended therapy considered.

TRAP A negative D-dimer only helps a low-probability patient. In pregnancy the pathway is different — chest X-ray and leg dopplers come first.

Sugar, potassium, sodium

Three emergencies where the sequence itself is the examined content, and where correcting too fast is its own wrong answer.

Diabetic ketoacidosis

Glucose >13.9 (or near-normal on an SGLT2 inhibitor), pH <7.3 , bicarbonate <18 , ketones ≥ 3.0 . HHS is the mirror image — glucose >33.3 , osmolality >320 , few ketones, fluids first and slower.

1 Fluid before insulin — 0.9% sodium chloride

The deficit is often 5–7 litres. Restore the circulation first; the insulin can wait the few minutes it takes.

2 Fixed-rate insulin at 0.1 units/kg/h

Keep the patient's usual long-acting basal insulin running alongside it. Do not stop it.

3 Potassium sets the pace

Below 3.5 → hold the insulin and replace first. Between 3.5 and 5.5 → 40 mmol/L in the bag. Above 5.5 → none, and recheck.

4 Add 10% glucose below 14, and treat to ketone clearance

Resolution is ketones <0.6 , pH >7.3 , bicarbonate >15 . Convert to subcutaneous only when eating, with an overlap.

TRAP The endpoint is ketone clearance, not the glucose. Bicarbonate is almost never the right option, and over-fast correction risks cerebral oedema.

Hyperkalaemia

Potassium above 6.0, or any level with ECG change — peaked T waves, flat or absent P waves, a widening QRS.

1 ECG first, immediately

Repeat the sample only if it costs no time. A haemolysed specimen is a real answer — but never when the ECG is abnormal.

2 ECG change or $K \geq 6.5$ → IV calcium gluconate now

10 mL of 10%, repeated if changes persist. It stabilises the myocardium; it does not lower the potassium at all.

3 Shift it inside the cells

10 units of soluble insulin in 25 g of glucose, plus nebulised salbutamol. Then watch the glucose for several hours.

4 Remove it, and stop what caused it

ACE inhibitor, ARB, spironolactone, NSAID, trimethoprim. Potassium binder for the sub-acute problem; dialysis if refractory or in established renal failure.

TRAP Calcium buys minutes, insulin-dextrose buys hours, only removal is definitive. A question that stops at insulin-dextrose has not finished treating the patient.

Hyponatraemia

Sodium below 135. Volume status and paired osmolalities decide the answer — the number alone decides nothing.

1 Severe symptoms → hypertonic 3% saline, whatever the cause

Seizure, marked drowsiness or a falling GCS. Give it as boluses and stop chasing the aetiology until the brain is safe.

2 Otherwise assess volume status and send the right tests

Paired serum and urine osmolality with a urine sodium, plus thyroid and cortisol if the picture fits.

3 Treat by compartment

Hypovolaemic with urine Na <20 → 0.9% saline. Euvolaemic, urine osm >100 and urine Na >30 → SIADH: restrict fluid, find the drug, lung, CNS or malignant cause. Hypervolaemic → treat the failing organ.

4 Cap the correction at 8–10 mmol/L in 24 hours

Lower still if malnourished, alcohol-dependent, hypokalaemic, or the sodium has been low for a long time.

TRAP Over-rapid correction causes osmotic demyelination. Exclude pseudohyponatraemia — high glucose, paraprotein or lipids — before treating anything.

Sepsis, bleeding, failing kidneys

The three acute-take pathways where a timing figure — one hour, twenty-four hours, forty-eight — is usually the marked point.

Sepsis

Infection plus organ dysfunction. qSOFA ≥ 2 — respiratory rate ≥ 22 , altered mentation, systolic ≤ 100 — is a prompt to act, not a diagnosis.

1 Recognise it and start the clock

Senior review early. Do not wait for a lactate result before beginning treatment.

2 Cultures, lactate, urine output — then oxygen, antibiotics, fluids

Blood cultures before the first dose, but only if that costs no delay. Broad-spectrum IV antibiotics within one hour.

3 30 mL/kg crystalloid if hypotensive or lactate ≥ 4 , then reassess

Reassess perfusion after each bolus rather than running fluid blindly.

4 MAP still under 65 → noradrenaline and critical care, and get source control

Drain the collection, remove the line, take them to theatre. Source control is what actually ends the sepsis.

TRAP Antibiotics within the hour beat a perfect microbiological work-up. Never delay the first dose for imaging or a second set of cultures.

Upper gastrointestinal bleed

Haematemesis, coffee-ground vomit or melaena. Chronic liver disease anywhere in the stem changes the whole answer.

1 Resuscitate and score

Two large-bore cannulae, crossmatch, Glasgow-Blatchford. A score of 0–1 in a well patient can be managed without admission.

2 Transfuse restrictively — target Hb 70 g/L, or 80 with cardiac disease

Over-transfusion raises portal pressure and makes variceal bleeding worse.

3 Suspected varices → terlipressin plus prophylactic antibiotics before endoscopy

For non-variceal bleeding, a PPI before endoscopy adds nothing — it belongs after the endoscopic diagnosis.

4 Endoscopy within 24 hours, within 12 if unstable or variceal

Correct coagulopathy, stop the anticoagulant, and reverse it when the bleeding is life-threatening.

TRAP Antibiotics in variceal bleeding are a mortality intervention, not a courtesy. A urea rise out of proportion to the creatinine is the classic upper GI clue.

Acute kidney injury

Creatinine $\times 1.5$ within 7 days, a rise of ≥ 26.5 $\mu\text{mol/L}$ within 48 hours, or urine output under 0.5 mL/kg/h for 6 hours.

1 Exclude obstruction before anything clever

Catheter and bladder scan; ultrasound within 24 hours when no cause is obvious.

2 Assess volume and perfusion honestly

Hypovolaemic → fluid challenge and reassess. Already overloaded → the answer is not more fluid.

3 Stop the nephrotoxins and re-dose everything else

NSAIDs, ACE inhibitors and ARBs, aminoglycosides, contrast. Review every drug for renal adjustment.

4 Dip the urine — it is the cheapest test on the sheet

Blood and protein without an obvious explanation → glomerulonephritis or vasculitis. Refer the same day.

DIALYSE FOR refractory hyperkalaemia, refractory acidosis, refractory fluid overload, uraemic pericarditis or encephalopathy, or a dialysable poison.

Oncological emergencies

Each clue line below is the phrase the stem will use. These five carry more marks than their share of the blueprint suggests, because every one of them is a timed decision.

Neutropenic sepsis

Fever ≥ 38.3 °C once, or ≥ 38.0 for an hour, within six weeks of chemotherapy.

- 1 **Antibiotics within one hour**
Piperacillin-tazobactam as the usual first line. Do not wait for the neutrophil count.
- 2 **Culture peripherally and from every line lumen**
Full septic screen, then the rest of the sepsis bundle as normal.
- 3 **Escalate or step down deliberately**
Vancomycin for line infection or MRSA risk. Score with MASCC — low-risk patients can move to oral therapy.

TRAP A normal white cell count on arrival excludes nothing. The first dose goes in before the FBC comes back.

Tumour lysis syndrome

Bulky chemo-sensitive tumour, days into treatment: high potassium and phosphate, high urate, low calcium, rising creatinine.

- 1 **Stratify risk before cycle one**
High-grade lymphoma, ALL, high white count, bulky disease, existing renal impairment.
- 2 **Fluids for everyone at risk, then urate control**
Rasburicase for high risk, allopurinol for lower risk.
- 3 **Treat the potassium first**
It kills before the urate does. Cardiac monitoring, and renal input early for refractory disturbance.

TRAP Rasburicase is contraindicated in G6PD deficiency, and calcium for asymptomatic hypocalcaemia risks precipitation with phosphate.

Hypercalcaemia of malignancy

Known cancer with confusion, constipation, polyuria and thirst, or a corrected calcium above 3.0 mmol/L.

- 1 **Correct for albumin before you react**
Measured calcium + $0.02 \times (40 - \text{albumin in g/L})$.
- 2 **IV 0.9% saline is the treatment, not the preparation**
Rehydration alone brings the calcium down substantially.
- 3 **Then a renally-adjusted IV bisphosphonate**
Zoledronic acid; the nadir takes 2–4 days. Denosumab if refractory or in renal failure.

TRAP Loop diuretics are not routine — only once fluid-replete and overloaded. Stop thiazides, calcium and vitamin D.

Metastatic cord compression

New or progressive back pain with leg weakness, a sensory level, or bladder and bowel change in a patient with cancer.

- 1 **Dexamethasone immediately**
Typically 16 mg daily with gastric protection — before the imaging is even booked.
- 2 **Whole-spine MRI within 24 hours**
Sooner if the neurology is progressing. Whole spine, because multiple levels are common.
- 3 **Same-day radiotherapy or surgical decision**
Nurse flat until the spine is declared stable.

TRAP Function at the moment of treatment predicts function afterwards. Once they cannot walk, they usually will not walk again.

Superior vena cava obstruction

Facial and arm swelling with distended neck and chest-wall veins, breathlessness worse on stooping — usually with a known or suspected intrathoracic primary.

- 1 **Assess the airway first — stridor is what makes this a true emergency**
Sit the patient up, give oxygen, and start dexamethasone.
- 2 **CT chest with contrast to define the level, the cause and any thrombus**
Get a tissue diagnosis where the patient can wait for it — the histology dictates the definitive treatment.
- 3 **Endovascular stenting for rapid relief, then treat the tumour**
Radiotherapy or chemotherapy as the histology directs.

TRAP In a chemo-sensitive tumour — small-cell, lymphoma, germ cell — systemic treatment relieves it quickly, so an undiagnosed patient needs a biopsy, not just a steroid.